

Korea Robotics and Humanoids

How we envisage the humanoid era; HMG robotics value at \$57bn; stay Overweight on HMC/Kia

We remain in the bull camp on humanoids, since: 1) early deployments can create a learning flywheel: putting robots into real workflows accelerates iteration; and 2) humanoid's stranded capex risk is structurally lower. We expect early US demand to outstrip supply: while task-specific robots often win on throughput per dollar in structured settings, humanoids should gain share in less structured work with frequent edge cases. Industrial adoption should lead given higher willingness to pay and clearer ROI, with consumer adoption inflecting later as BOM declines. We believe the key bottlenecks are brain ('data'), hands, and supply chain readiness; we view Hyundai Motor Group (HMG) checks two of the three key boxes, namely 1) access to proprietary data; and 2) supply chain readiness, while early deployments can rely on simple grippers, while dexterous hands can be acquired or outsourced later. We retain Overweight on Hyundai Motor (HMC)/Kia and prefer robot OEMs (HMC/Kia) over parts suppliers (Mobis, Mando, Autoveer) as value capture shifts toward system integrators.

- Humanoids: deploy early, learn faster, without locking in obsolete capex.** While the timing of a true 'breakthrough moment' remains uncertain, we remain in the bull-camp for humanoids for two reasons: 1) early deployments can create a data and learning flywheel: the sooner robots are placed in real environments, the faster performance can improve through iteration; and 2) the 'stranded capex risk' is lower than many assume because humanoids are upgradeable, reducing the risk of stranded capex if tasks evolve. Data security issues, often cited as a key adoption hurdle for many customers, can be mitigated through on-premise deployments and robust data governance frameworks that limit the risk of working data leakage.
- We do not underestimate the advantages of task-specific robots; yet we expect early US humanoid demand to outstrip supply.** End-customers would not buy humanoids per se, but they will buy automation that improves throughput, safety, and total cost. In environments with clear procedures and limited edge cases, task-specific robots should often deliver better throughput and accuracy per dollar than humanoids. Over time, however, we expect humanoids to gain share in less structured work where procedures are variable and edge cases are frequent, which is common in real operations. Ultimately, the highest-ROI setup may be a hybrid deployment where humanoids and task-specific robots are co-located, so one form factor is not forced to do everything. Taking all these into consideration, we estimate US humanoid demand could reach ~531k units by 2030 (~306k units excluding consumer), exceeding our production capacity forecast of ~300k units.
- Industrial scale first, before consumer adoption.** We expect the humanoid market to develop first in industrial applications, where willingness to pay is higher and ROI is clearer, given today's high BOM costs and limited consumer

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Equity Ratings and Price Targets

Company	Ticker	Mkt Cap (\$ mn)	Price CCY	Price	Rating		Price Target		
					Cur	Prev	Cur	End Date	Prev End Date
Hyundai Motor Company	005380 KS	77,572	KRW	413,000	OW	n/c	540,000	Dec-27	530,000 n/c
Kia Corp	000270 KS	37,273	KRW	131,200	OW	n/c	210,000	Dec-27	200,000 n/c

Source: Company data, Bloomberg Finance L.P., J.P. Morgan estimates. n/c = no change. All prices as of 24 Aug 26.

See page 32 for analyst certification and important disclosures, including non-US analyst disclosures.

affordability. Industrial-first players can generate revenue while building real-world data, reliability, and manufacturing expertise, with scale driving cost reductions. As hardware components and AI models become more standardized, consumer-side barriers to entry should fall through lower BOM, and then the much larger consumer addressable market can inflect. We believe the strongest strategy is to design humanoids to be consumer-compatible from the outset, but monetize and validate them in industrial settings first, until safety and reliability are proven at scale and BOM make consumer pricing viable. We believe HMG is broadly following this playbook.

- **Three bottlenecks define the pace of humanoid scale-up: brain, hands, and supply chain readiness.** We see the largest bottleneck in the brain (especially, quality training data). Dexterous hands are another major hurdle; we expect many early deployments to use simpler grippers rather than fully dexterous hands, prioritizing reliability over versatility. Supply chain readiness will also matter, both in the ability to deliver customized components on time and balancing efficiency vs. sovereign control (i.e., diversifying parts exposure beyond China where needed). We believe HMG checks two of the three key boxes, namely 1) access to proprietary data; and 2) supply chain readiness. Dexterous hands can likely be acquired or outsourced over time, in our view. Importantly, HMG has time to wait for the right hands partners to emerge, as its initial humanoid deployment targets are not highly dexterous tasks (e.g., machine tending).
- **Robotics upside remains valid; stay Overweight on HMC/Kia and prefer robot OEMs over parts suppliers.** We refresh our humanoid ASP framework and continue to view P/S as the more relevant valuation method at this stage, as early deployments may price close to COGS and remain loss-making during the ramp. On this basis, we estimate the robot business value at ~\$57bn, up 16% vs. our earlier work in January 2026 (~\$49bn). Across the broader sector, we continue to prefer OEMs over parts suppliers, given OEMs' increasing discretion across the value chain in both autos and robotics. In our [latest note](#), we highlighted HMC/Kia's US HEV-led M/S gains and argued that the bulk of Street auto earnings downgrades are likely behind us. As the market appears to have found an earnings bottom, we expect investor focus to shift toward the longer-term narrative, particularly tangible progress in robotics.

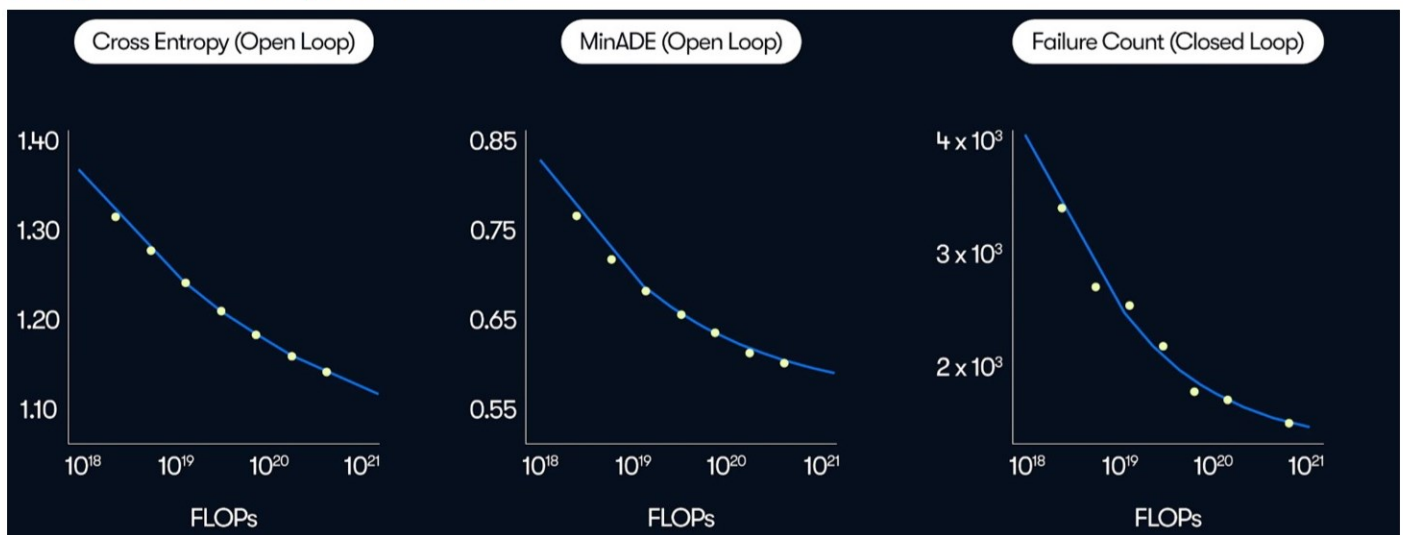
Humanoids: deploy early, learn faster, without locking in obsolete capex

While the timing of a true ‘breakthrough moment’ remains uncertain, we remain in the bull-camp for humanoids for two reasons: 1) early deployments can create a data and learning flywheel: the sooner robots are placed in real environments, the faster performance can improve through iteration; and 2) the ‘stranded capex risk’ is lower than many assume because humanoids are upgradeable, reducing the risk of stranded capex if tasks evolve. Data security issues, often cited as a key adoption hurdle for many customers, can be mitigated through on-premise deployments and robust data governance frameworks that limit the risk of working data leakage.

Early deployment can create a data and learnings flywheel

Early deployments can create a data and learning flywheel. More real-world data helps humanoids adapt to edge cases over time, similar to autonomous driving where greater mileage and edge-case exposure improves model accuracy. In this context, earlier development and deployment into real work sites can create a moat, as first movers accumulate real-world edge-case data sooner and build more robust ‘instincts’ into their control stack.

Figure 1: Motion forecasting quality follows a power-law as a function of training compute; self-driving systems require vast datasets, the challenge lies in the diminishing returns of data input



Note: The objective the model is trained with (Left), motion forecasting open-loop metrics (Middle), and planning performance in closed-loop simulation (Right).
 Source: Waymo

Humanoids are upgradeable; ‘stranded capex risk’ is lower

Humanoids are built with redundancy, which can translate into higher upfront costs. However, that redundancy can also reduce ‘stranded capex’ risk for customers. If a task becomes obsolete or demand shifts, humanoids can often be redeployed to new roles with relatively minor software updates and workflow adjustments, materially lowering the risk of owning idle, single-purpose automation.

Market-leading humanoids will be deployed everywhere: there will be no concerns of working data leakage

We believe humanoid adoption is less constrained by competitive conflicts of interest than commonly assumed. For example, we think Toyota could deploy HMG humanoids without outsized data-leakage concerns for two reasons: 1) sensitive operational data can be converted into abstracted motion vectors that are not easily reverse-engineered into proprietary processes; and, more importantly, 2) data can be stored and processed in customer-owned ‘on-premise datacenters’, limiting robot OEM access. At the end of the day, we expect **tiered subscription structures** to balance data sharing preferences, OTA update speed, and subscription pricing.

Potential conflicts can also be managed through tiered subscription structures

One model could be: 1) customers that allow full data sharing (including the most raw, first-layer data such as video) in exchange for the lowest subscription cost and fastest OTA updates; 2) customers that share only vectorized/limited telemetry data, paying a mid-tier fee while still receiving timely OTA updates; and 3) customers that keep data fully on-prem with minimal or no data outflow, paying a higher subscription fee and potentially additional charges for customized OTA support.

We do not underestimate the advantages of task-specific robots; yet we expect early US humanoid demand to outstrip supply

End-customers would not buy humanoids per se, but they will buy automation that improves throughput, safety, and total cost. In environments with clear procedures and limited edge cases, task-specific robots should often deliver better throughput and accuracy per dollar than humanoids. Over time, however, we expect humanoids to gain share in less structured work where procedures are variable and edge cases are frequent, which is common in real operations. Ultimately, the highest-ROI setup may be a hybrid deployment where humanoids and task-specific robots are co-located, so one form factor is not forced to do everything. Taking all these into consideration, we estimate US humanoid demand could reach ~531k units by 2030 (~306k units excluding consumer), exceeding our production capacity forecast of ~300k units.

We think co-locating humanoids and task-specific robots can deliver the highest-ROI pattern, because you stop forcing one form factor to do everything. The trick is to partition the work: task-specific robots handle the repeatable, high-throughput, high-reliability 'core loop', while humanoids handle the messy edges: tight spaces, human-facing interaction, ad-hoc manipulation, and rapid task switching. A practical example is last-mile delivery, once introduced by HMG: Stretch (one-armed logistics robot) provides high-throughput box handling at the vehicle interface, while the Atlas humanoid handles the irregular 'last-meter' tasks around people, doors, and curves.

Table 1: Humanoids vs. Special-purpose robots: a practical fit guide (when each wins)

	Humanoids fit better in...	Task-specific robots prosper in...
Process	Steps aren't fully defined; lots of "do what you see" work	Clear "manuals" exist: steps are clear, stable, and easy to program
Edge cases	Many weird exceptions; environment changes often	Few exceptions; you can control the environment
Value realization	Value comes from doing many tasks over time	Value is immediate from one high-volume task
Throughput	Speed matters less than flexibility	Speed/repeatability is the main goal
Task variety	Many different tasks per shift	One/few tasks repeated at scale
Workspace	Human spaces (mixed objects, clutter, tight areas)	Designed cells/lanes (fixtures, conveyors, marked bins)
Operations	You can iterate (pilot > improve > expand)	You need deterministic performance day 1

Source: J.P. Morgan

Figure 2: Humanoids can handle true last-mile tasks and navigate messier, edge-case-heavy environments



Source: Kia

Figure 3: Diverse robots can be co-located to deliver the highest ROI



Source: Kia

Early-stage demand in the US would exceed supply

The US humanoid demand outlook is fundamentally anchored in structural labor shortages rather than outright labor replacement. Based on [our US industrial team's analysis](#), the US manufacturing industry currently faces ~462k unfilled jobs, projected to reach ~1.6 million by 2030, with an estimated 25% of the current gap sitting in humanoid-addressable roles (material handling, parts transfer, machine tending, and quality inspection), expanding to 50% of the gap by 2030. In warehousing, humanoids operating at ~\$10-12/hour vs. ~\$30/hour for human labor create an addressable pool of ~800k roles in 2026 growing to ~1.2 million by 2030, with penetration ramping from 1% to 16%. Separately, the consumer segment offers the greatest long-term optionality, scaling from ~10k units towards ~225k units (~0.5% of >\$100k households).

Assuming humanoids will serve as a complement to human labor rather than a substitute, with full-time equivalent coverage currently requiring ~2 humanoids per human worker before improving toward 1.2-1.3 human equivalency by 2030, total US demand will reach ~531k units by 2030 (303k units ex-consumer), exceeding our ~300k production capacity forecast.

Figure 4: US humanoids demand landscape

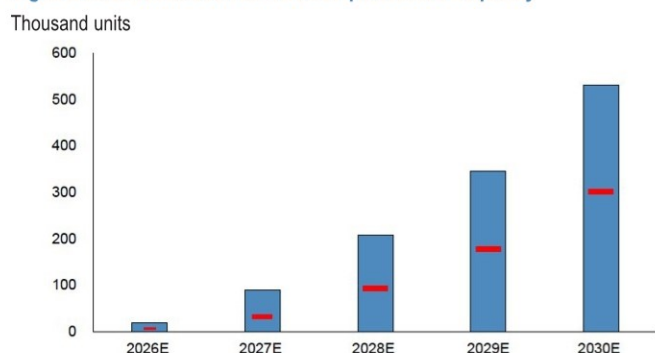
'000 jobs

	2026E	2027E	2028E	2029E	2030E
Manufacturing					
Manufacturing Labor Opportunity	462	747	1,031	1,316	1,600
% of jobs addressable	25%	35%	40%	45%	50%
FTE coverage conversion	2.1	1.8	1.7	1.4	1.3
Penetration	2%	5%	10%	15%	20%
Manufacturing Unit Demand	4,851	23,776	68,871	126,979	200,000
YoY		390%	190%	84%	58%
Logistics & Warehousing					
Logistics & Warehousing Labor Opportunity	800	900	1,000	1,100	1,200
% of jobs addressable	25%	30%	35%	40%	45%
FTE coverage conversion	2.0	1.7	1.4	1.3	1.2
Penetration	1%	3%	7%	11%	16%
Logistics & Warehousing Unit Demand	4,000	13,978	35,035	64,372	101,952
YoY		249%	151%	84%	58%
Consumer					
Total U.S. Households (k)	133,675	135,413	137,173	138,956	140,763
YoY Growth		1.3%	1.3%	1.3%	1.3%
% of households >\$100k	30%	30%	30%	31%	32%
Consumer - eligible households	40,000	41,000	41,000	43,000	45,000
Penetration	0.03%	0.13%	0.25%	0.35%	0.50%
Units per household	1	1	1	1	1
Consumer Unit Demand	10,000	51,250	102,500	150,500	225,000
YoY		413%	100%	47%	50%
Healthcare, Defense, & Other					
Unit Demand	35	407	1,189	2,658	4,519
YoY		1064%	192%	124%	70%
U.S. Unit Demand Across End Markets	18,886	89,411	207,595	344,508	531,471
YoY		373%	132%	66%	54%
U.S. % of production	299%	279%	226%	194%	177%

Note: FTE = Full Time Employee Equivalent

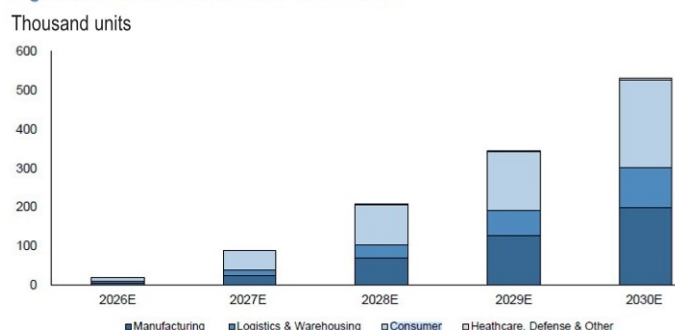
Source: J.P. Morgan estimates

Figure 5: US humanoid demand vs. production capacity



Note: Red = estimated US Humanoid Robot Units Production
Source: J.P. Morgan Estimates

Figure 6: End market demand breakdown



Source: J.P. Morgan estimates

Industrial scale first, before consumer adoption

We expect the humanoid market to develop first in industrial applications, where willingness to pay is higher and ROI is clearer, given today's high BOM costs and limited consumer affordability. Industrial-first players can generate revenue while building real-world data, reliability, and manufacturing expertise, with scale driving cost reductions. As hardware components and AI models become more standardized, consumer-side barriers to entry should fall through lower BOM, and then the much larger consumer addressable market can inflect. We believe the strongest strategy is to design humanoids to be consumer-compatible from the outset, but monetize and validate them in industrial settings first, until safety and reliability are proven at scale and BOM make consumer pricing viable. We believe HMG is broadly following this playbook.

Two-stage adoption path: Stage 1 industrial scale, then Stage 2 consumer proliferation

The key hurdles separating the two stages are:

- **Economics:** Consumer willingness to pay is likely anchored in the hundreds of dollars per month range (for example, industry commentary ~\$500/month from Figure AI and 1X Technology), while current general-purpose humanoid BOMs remain in the tens of thousands to >\$100k, making consumer payback unattractive vs. commercial use cases.
- **Non-economic barriers:** Consumers require proven safety, trust, and productization in uncontrolled home environments (kids, pets, unpredictable edge cases), alongside evolving standards and certification.
- **The reality is that 'digital AI' already covers many household tasks without a humanoid body,** leaving physical dexterity and manipulation as the key incremental use case for humanoids.

Strategy implication: The practical playbook is to build a consumer-compatible platform, then harden it in industrial settings until: 1) safety and reliability are proven at scale; and 2) BOM and service economics make consumer pricing viable.

Why HMG is directionally well positioned

We believe HMG is broadly following the playbook discussed above. HMG has a built-in Stage 1 environment in the form of large factories and repeatable industrial workflows, which can justify early deployments while BOM remains high and generate meaningful operational learning. In our view, HMG's advantage is the combination of clearer industrial ROI and higher willingness to pay (from internal demand and potential external partners), auto-style supply chain leverage and standardization to move down the cost curve, and factory deployments that act as a "data and reliability engine" to improve uptime, safety protocols, and service processes ahead of broader consumer rollout.

Three bottlenecks define the pace of humanoid scale-up: brain, hands, and supply chain readiness

We see the largest bottleneck in ‘brain’ (especially, quality training data). Dexterous hands are another major hurdle; we expect many early deployments to use simpler grippers rather than fully dexterous hands, prioritizing reliability over versatility. Supply chain readiness will also matter, both in the ability to deliver customized components on time and balancing efficiency vs. sovereign control (i.e., diversifying parts exposure beyond China where needed). **We believe HMG checks two of the three key boxes, namely: 1) access to proprietary data; and 2) supply chain readiness.** Dexterous hands can likely be acquired or outsourced over time, in our view. Importantly, HMG has time to wait for the right hands partners to emerge, as its initial humanoid deployment targets are not highly dexterous tasks (e.g., machine tending).

1. ‘Good data’ is all you need

A robot foundation model is a large, general-purpose AI model pretrained on diverse robotic and multimodal data to perform and generalize across multiple tasks, environments, and robot embodiments. We note that there are already multiple robot foundation model options (indicated in Table 2), with more in development, and we expect leading models to become broadly available across multiple hardware ecosystems over time. Instead, **we believe data, especially high-quality ‘good data’ used for training and adaptation, will be a primary driver of real-world performance differentiation.**

There are multiple options for robot foundation models

There are multiple options for robot foundation models, many of which are open-source or open-weight (publicly downloadable, but without full access to the source code), allowing users to fine-tune the models for customization. We highlight representative models in Table 2, but note that more are in development, which should expand robot OEMs’ choices to select and tailor the most suitable model for their platforms.

Table 2: Robot foundation models

Robot foundation model	Developer	Representative benchmark (LIBERO*)	Model availability
GR00T N1.x	NVIDIA	96.5%	Open-weight
Gemini Robotics	Google DeepMind	n/a	Closed
π 0.5	Physical Intelligence	96.85-96.95%	Open-weight
ACT-1	Sunday Robotics	n/a	Closed
OpenVLA-OFT	Stanford/ UC Berkeley	97.1%	Open-weight
SmoVLA	Hugging Face	88.8%	Open-source
OpenHelix	Zhejiang Univ./ Westlake, et al.	n/a	Open-source
X-VLA	Tsinghua/ Shanghai AI Lab, et al.	98.1%	Open-source

*LIBERO: A widely used simulation benchmark for evaluating robot manipulation and VLA models across 40 core tasks, testing spatial understanding, object manipulation, goal following, and long-horizon execution.

Source: Company data, J.P. Morgan

What defines ‘good data’?

Good data is data that links what the robot sensed to what the robot did, and what happened next, under realistic variability.

As much multimodal data as possible helps: not only visual images, but also tactile (touch), force/torque, proprioception (joint position/velocity/effort), audio, and even thermal signals can constitute ‘good data’, especially for manipulation. This kind of multimodal data helps close the ‘sim-to-real gap’.

- **Definition: sim-to-real gap.** The sim-to-real gap is the difference between what happens in simulation and what happens in the real world, due to imperfect physics, unmodeled friction/compliance, sensor noise, latency, contact dynamics, material properties, and environment variability. When that gap is large, policies that look great in simulation can perform poorly or fail in reality.

Large-motion tasks such as walking, running, dancing, or simply pushing tend to transfer well from virtual simulation because the real-world interaction is relatively simple: contact is mostly feet-to-ground, precision tolerances are looser, and performance is less sensitive to friction details, so physics simulation is comparatively accurate (a small sim-to-real gap). On the other hand, **Dexterous manipulation** transfers are less accurate because the interaction is inherently contact-rich and high-precision: multiple fingers touch objects at once, millimeter-level errors matter, outcomes depend strongly on friction and deformation, and tactile/force feedback is often essential. As a result, small simulation inaccuracies compound quickly during dexterous actions, creating a much larger sim-to-real gap and frequent behavioral mismatch.

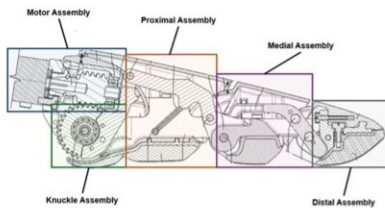
Therefore, good data that fills the sim-to-real gap has the following characteristics:

- **Multimodal synchronization:** All sensor streams are time-aligned (e.g., camera frames line up with tactile events and joint torques). This is crucial because manipulation depends on when contact occurred, when slip started, and how the controller responded.
- **Precise action-observation pairs:** For every moment, you know exactly what the robot perceived (camera/tactile/proprioception/FT) and exactly what command it executed (joint targets/torques/gripper forces), with accurate timestamps. If the timing or labeling is off, the model learns the wrong cause-and-effect.
- **Diversification:** Demonstrations cover a real-world variety of different objects, colors, textures, sizes, lighting, backgrounds, table heights, robot base poses, grasp points, approach angles, and different human demonstrators/robot embodiments. The goal is to prevent “overfitting to one lab setup.”
- **Contact-rich signals:** The dataset includes what matters during touch: contact onset, normal force, shear force, torque, pressure distribution, finger pose, micro-slip indicators, and how those evolve during grasp/lift/rotate/place. This is the core missing ingredient in many vision-only datasets.
- **Failure + recovery trajectories:** Not only success demos. You should also include drops, slips, missed grasps, misalignments, and most importantly, how the policy recovers (re-grasp, reposition, reduce force, change approach). Recovery data is often what turns a “demo robot” into a “production robot.”

2. Dexterous hands

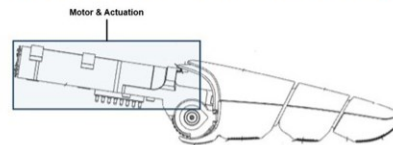
Based on our interviews with corporates and robotics experts, hand manipulation remains one of the most technically challenging parts of humanoids. The human body has ~70-100 DoFs (degrees of freedom), while two hands alone account for ~40-54 DoFs (~20-27 each): hands are small, but the actuators and sensors must be densely packed to achieve dexterity. As shown in Figures 7-10, the hands and fingers are particularly challenging, both from a component sourcing perspective and in terms of assembly. As a result, hands are mechanically more difficult to design, while VLA models that drive dexterous hands must process more complex information and longer action sequences.

Figure 7: Component breakdown of a finger



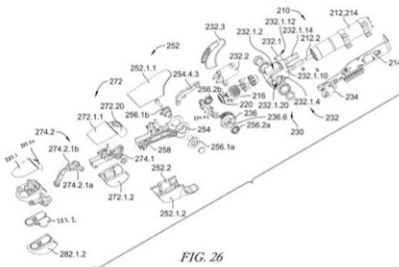
Note: Figure 23A of US 2025/0205908 A1 Patent published on June 26, 2025
Source: Company reports, Google Patents, J.P. Morgan

Figure 8: Component breakdown of a finger



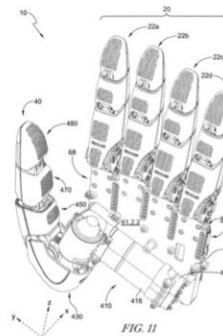
Note: Figure 17 of US 2025/0205908 A1 Patent published on June 26, 2025
Source: Company reports, Google Patents, J.P. Morgan

Figure 9: Breakdown of components within a finger



Note: Figure 26 of US 2025/0205908 A1 Patent published on June 26, 2025
Source: Company reports, Google Patents

Figure 10: Overview of a humanoid hand



Note: Figure 11 of US 2025/0205908 A1 Patent published on June 26, 2025
Source: Company reports, Google Patents

We see many dexterous hand products textualized in Table 3, but none of them match human performance. There are fundamental trade-offs: 1) more dexterity typically requires more actuators and sensors, increasing weight (which then requires more torque and more energy); 2) using lighter actuators can help, but often increases cost and leads to poor heat dissipation; and 3) smaller, denser mechanical parts can increase failure rates, and repairs become harder.

In the early stages of humanoids (before ~2030), we think deployments can largely sidestep the fully dexterous hand bottleneck: companies can target jobs that work with simple grippers first, and develop or outsource advanced hands later.

Table 3: Humanoid hand comparison vs. human hand

Product	Company	Mechanism	Price (\$)	Finger count	Kapandji score	DOF	Backdrivability	Abduction	Tactile/palm	Tactile/fingers	Hand length (mm)	Weight (g)	Fingertip force (N)	Rated load (kg)
Agibot Omnihand Pro	Agibot	Tendon-driven	~14,610	5	10/11	12 active/ 19 total	No	Yes	Yes	Yes	207	750	20	35*
XHAND 1	Robotera	In-finger actuator	~14,500	5	8/11	12	Yes	Yes	No	Yes	191	1,100	15	n/a
Allegro V5 Plus	Wonik Robotics	In-finger actuator	~11,000	4	9/11	16	Yes	No	No	Yes	255	1,024	n/a	15
LEAP Hand V1	Carnegie Mellon Univ.	Tendon-driven	~2,000	4	9/11	16	Yes	Yes	No	No	232	595	n/a	n/a
Linkerbot L20	Linkerbot	In-finger actuator	~12,200	5	8/11	16	No	Yes	Yes	Yes	264	1,000	10	5
Orca Hand V1	ETH Zurich	Tendon-driven	~7,600	5	9/11	16	Yes	Yes	No	No	330	1,100	n/a	2
DG-SF-M	Tesollo	In-finger actuator	~21,000	5	10/11	20	Yes	Yes	No	Yes	n/a	1,763	97	12
DG-SF-S	Tesollo	In-finger actuator	~21,000	5	10/11	20	Yes	Yes	No	Yes	210	880	19	12
Robotis HX5-D20	Robotis	In-finger actuator	~6,699	5	11/11	20	Yes	Yes	No	Yes	n/a	1,360	14	15
Wuji Hand V1.1	Wuji Tech	In-finger actuator	~16,000	5	10/11	20	No	Yes	No	No	201	590	15	20
Figure 03 Hand	Figure AI	In-finger actuator	n/a	5	n/a	20	Yes	Yes	Yes	Yes	n/a	n/a	n/a	20
Brainco Revo3	BrainCo	In-finger actuator	~8,000	5	11/11	21	Yes	Yes	Yes	Yes	n/a	n/a	20	n/a
Sharpa Wave	Sharpa Robotics	In-finger actuator	~52,000	5	9/11	22	Yes	Yes	No	Yes	200	1,300	n/a	15
Optimus V3 Hand	Tesla	Tendon-driven	n/a	5	n/a	22	Yes	Yes	Yes	Yes	n/a	n/a	n/a	9
Unitree Dex5-1	Unitree	In-finger actuator	~23,500	5	n/a	16 active/ 20 total	Yes	Yes	Yes	Yes	217	1,100	10	5
1X NEO Hand	1X Technologies	Tendon-driven	~20,000	5	n/a	22	Yes	Yes	Yes	Yes	n/a	n/a	45	9
Human hand (indicative only)				5	11/11	20-27	Yes	Yes	Yes	Yes	400-500		10-50	30-50

Kapandji score quantifies how well the thumb can reach the other fingers and palm to assess overall hand dexterity.

Backdrivability is the ability of an actuator or joint to be moved by an external force.

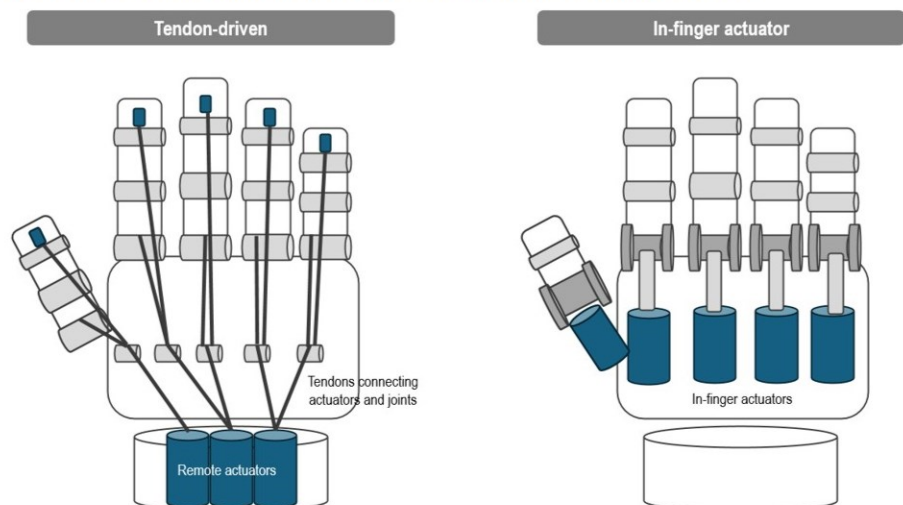
Abduction refers to the ability of the fingers to spread sideways, improving grasp versatility across different object shapes.

Source: Company data, J.P. Morgan estimates

Tendon-driven structure vs. in-finger actuator structure

- A tendon-driven hand places actuators away from the fingers (often in the forearm/ wrist) and transmits force through tendons (cables) to finger joints, similar to how human tendons route muscle force. The advantages of this design are potentially higher dexterity, lighter fingers, and human-like compliance. On the other hand, key shortcomings are that they are harder to manufacture and calibrate (tendon routing is complex), maintenance can be difficult, and replacement often requires disassembly of the wrist/forearm.
- An in-finger actuator hand places actuators closer to the joints they move (in the palm/fingers), reducing tendon complexity. Key advantages of this design include simpler control and assembly. The disadvantages are relatively limited dexterity, heavier fingers/palm that leads to heat and power density challenges.

Figure 11: Robot hand comparison in image: tendon-driven vs. in-finger actuator



Source: J.P. Morgan

3. Supply chain readiness

Even if the ‘brain’ (software/sufficiently good training data) is ready and advanced hands are developed in-house or outsourced, humanoids cannot scale to thousands or millions of units without a stable, high quality supply chain ecosystem.

Compared to conventional industrial robots, which typically use ~2,000 parts, humanoids require many more parts (~5,000 parts), including more sensors, computing, power electronics, actuators, and complex mechanical subassemblies. Auto OEMs have long managed extremely complex ecosystems, historically ~30,000 parts for ICE vehicles and ~15,000 parts for EVs. We also see meaningful overlap (>70% based on our global Auto team estimate) between humanoids and EV supply chains, which includes: 1) motors and power electronics; 2) battery systems (batteries, BMS, converters); 3) cameras and vision systems.

Therefore, we believe that the auto OEMs' expertise in supplier management, quality control, and scaling complex electromechanical systems can be leveraged meaningfully in humanoids, particularly for building a reliable ecosystem that can support high volume production.

Figure 12: Areas of Overlap: Humanoids vs. EVs/Autonomous Vehicles

Subsystem	Representative EV Components	Representative Humanoid Components
Batteries and Power Systems	400–800V packs; module/pack BMS; liquid cooling plates; HV contactors; DC-DC converters	48–52V packs; compact BMS; heat spreaders; quick swap/back mounted packs
Computing Power Chips	AV/ADAS compute (SoCs/GPUs), safety MCUs, domain controllers, vehicle Ethernet	Head/torso AI module, gait controller, arm/hand controllers, joint MCUs, safety supervisor
Cameras and Vision Systems	Front/rear/side cameras; surround view; HDR sensors; heated housings	Head-mounted stereo; chest RGBD; wrist cameras for grasping; protective domes
Sensors	6-axis IMUs; short range ultrasonics; 77/79 GHz radar; load cells	High-grade torso IMU; foot force/torque; fingertip tactiles; proximity sensors on limbs
LiDAR	Roof/fender solid state LiDAR; long range units	Head or shoulder mounted solid-state LiDAR; short to mid range 360° mapping
Motors, Actuators, and Power Electronics	Traction motors; e axle inverters; EPS/steer actuators; DC-link caps; gate drivers	Joint BLDCs with harmonic drives; shoulder/hip torque dense actuators; miniature inverters; brake/clutch modules
Materials, Structures, and Manufacturing	Gigacastings, extruded crash beams, sealed enclosures, automotive connectors	Torso/limb frames, swappable panels, sealed joint housings, quick-service connectors
Software, AI, and Tooling	AV simulators, OTA, safety monitors, telematics, data labeling tools	Robot simulators (locomotion/manipulation), OTA for control stacks, telemetry dashboards, safety supervisors

Source: J.P. Morgan ([From Autonomous Cars to Humanoids: Automotive industry embraces Humanoids](#))

HMG checks two of the three key boxes: 1) access to proprietary data; and 2) supply chain readiness

HMG's robot training center, RMAC (Robot Metaplant Application Center), opened this summer and has begun collecting real-world data through a demo factory environment. We believe RMAC will be central to building HMG's data flywheel by replicating workflows across multiple verticals where the Group has domain exposure, including autos, machinery, logistics, and construction. In parallel, Figure AI already released factory video clips, which helped convince investors that humanoids can be deployable in real industrial settings.

Separately, [news reports](#) indicate HMG requested long-term business plan updates from its auto parts suppliers as part of discussions on how the future supply chain should be structured. This suggests HMG intends to go beyond a simple buyer-supplier relationship and provide broader support, including technical assistance and robotics-related capability building, to actively guide suppliers' transition toward the robotics industry.

Robotics upside remains valid; stay Overweight on HMC/Kia

We refresh our humanoid ASP framework and continue to view P/S as the more relevant valuation method at this stage, as early deployments may price close to COGS and remain loss-making during the ramp. On this basis, we estimate the robot business value at ~\$57bn, up 16% vs. our earlier work in January 2026 (~\$49bn). Across the broader sector, we continue to prefer OEMs over parts suppliers, given OEMs' increasing discretion across the value chain in both autos and robotics. In our [latest note](#), we highlighted HMC/Kia's US HEV-led M/S gains and argued that the bulk of Street auto earnings downgrades are likely behind us. As the market appears to have found an earnings bottom, we expect investor focus to shift toward the longer-term narrative, particularly tangible progress in robotics.

Refreshing our assumptions on HMG humanoid's ASP and

Humanoid ASP: willingness to pay as the anchor

We believe an ASP of ~\$120k/unit remains a reasonable starting point in 2028-30E, driven by the industrial sector's willingness to pay rather than a cost-based BOM build-up. With US manufacturing labor costs around ~\$30/hour and the minimum wage standing at \$7-17/hour, we believe the economics can work. At ~\$120k ASP, assuming 20 hours/day uptime and a two-year payback horizon, the implied cost is ~\$8.2/hour. Electricity costs are negligible in this context: even at ~20kWh/day (assuming continuous heavy lifting), it is about ~\$0.1/hour at typical US power prices.

The ~\$120k/unit price point is supported by industry cross checks. We also look to external reference points: Figure AI's CEO, Brett Adcock, has suggested consumer subscription pricing around ~\$500/month and commercial pricing potentially ~10x higher. While not formal guidance, it is a useful indication of how leading management teams frame industrial customers' willingness to pay. At a \$5,000/month subscription, the implied cost over a two-year horizon (a common payback threshold in the industrial sector) is \$120k/unit.

- **RaaS (Robots-as-a-Service) as an alternative pricing model.** Humanoid OEMs can accelerate adoption by offering RaaS, where customers subscribe to robots for a recurring fee, typically with deployment, maintenance, updates, monitoring, and support bundled in, rather than purchasing humanoids upfront. For customers, RaaS lowers adoption friction by reducing upfront capex and implementation burden, enabling them to start small, prove ROI faster, and scale deployments up or down as needs change. For OEMs, RaaS can be attractive by building a recurring revenue stream, expanding the installed base, and improving customer retention.

Ultra-low ASP is unlikely to be the sole deciding factor for industrial humanoids.

Some humanoid developers (e.g., Tesla and select Chinese players) have floated long-term targets of \$20-30k/unit, in theory achievable through aggressive BOM compression and supplier cost-down. We believe industrial buyers will continue to prioritize ROI reliability over headline price, and will require robust uptime, service and repair coverage, and continuous performance improvement via OTA updates. As a result, an ultra-low ASP is unlikely to be the sole deciding factor, particularly in early industrial deployments.

Table 4: Humanoid ASP at \$120k, translating to \$8.2 hourly cost, appears competitive vs. US minimum wage and average manufacturing production worker's hourly wage

\$	
Humanoid ASP	120,000
Hourly cost (2-year deployment, 20 operating hours/day)	8.2
US minimum wage (hourly)	~7-17, depending on states
Average US manufacturing production worker hourly wage	~30

Source: J.P. Morgan estimates

Humanoid BOM likely ~\$53-63k/unit (thousands scale); hands are the key swing factor

Based on our additional channel checks, our [global Auto team's view](#), and improving supply chain readiness, we estimate HMG's initial production BOM in 2028-30E at ~\$53-63k/unit, with the largest sensitivity being 'hands'. We assume outsourcing at ~\$5-10k per hand, but higher-dexterity options (closer to human hands with 20+ DoF) are typically priced above ~\$20k, and the widely cited *Sharpa* hand can be ~\$52k/unit. If HMG opts for simpler grippers (2-3 digits) in early deployments, BOM could screen much lower.

Beyond the initial 30k units/year target, we expect further BOM compression, similar to auto supply chains. Our global Auto head, Jose Asumendi, expects a ~60-70% reduction in BOM over time, implying ~\$53-63k could decline to ~\$21-25k/unit. This would be driven not only by supply chain cost-down, but also by component de-specing as software compensates, and potential removal of certain sensors deemed non-essential (e.g., LiDAR and some force/torque sensors). Still, the sourcing of reasonably priced dexterous hands remains a key determinant of how fast BOM can fall.

Table 5: Humanoid initial production BOM vs. Scaled production BOM

Category	Sub-components	Quantity	Est. unit cost	Total cost	% of BOM (ex-hands)
Actuation (ex-hands)	Actuators (includes motor, reducer, controller, encoder)	30	700	21,000	49%
Dexterous hand modules		2	5,000 - 10,000	10,000 - 20,000	
Compute	NVIDIA Jetson AGX Thor	1	3,000	3,000	7%
Vision	Depth cameras + Modules	8	300	2,400	6%
	LiDAR (optional)	1	500	500	1%
Sensing	IMUs (Inertial Measurement Unit): torso, upper torso, head	3	100	300	1%
	6-axis F/T (force/torque) sensors: wrists, ankles (optional)	4	1,000	4,000	9%
Power	2.5kwh Li-ion battery, BMS, DC/DC converter, charging interface	1	3,000	3,000	7%
Electrical & Connectivity	Wiring harnesses, Communication board, I/O switches, Cooling	Set		4,000	9%
Structure	Frame, Joints, Shell (machined/3D)	Set		5,000	12%
Initial production BOM				53,200 - 63,200	
Scaled production BOM				~21,000 - 25,000	~60-70% cost reductions through volume ramp, design optimization

Source: Company data, J.P. Morgan estimates ([From Autonomous Cars to Humanoids: Automotive industry embraces Humanoids](#))

BOM itself may not be the binding constraint; a larger hurdle for commercialization would be the overhead required to build real-world data

We do not view BOM as the primary constraint, assuming OEMs can secure capable vendors, although initial BOM levels may not be optimized for lowest cost. As volumes scale and component standardization increases, we expect OEMs to drive cost reduction

through further parts commonization and selective, flexible use of Chinese suppliers where feasible.

Instead, we see overhead costs as a bigger hurdle for accumulating real-world data, including: 1) datacenter infrastructure; 2) demo-plant setups that replicate real workflows; 3) internal robot fleets used for training rather than customer sales; 4) a dedicated workforce for data collection; and 5) teleoperation and data-capture equipment. The most direct way to reduce this burden is to scale volumes.

HMG robotics business value and potential further upside

Valuation

We assign a 20x P/S multiple on 2030E revenue, when we assume HMG will deliver a full-scale humanoid production run-rate of 30k units/year, is reasonable, considering:

- While comparable examples are limited, ex-China robotics names often trade at higher multiples, reflecting earlier-stage growth profiles and, in many cases, higher monetization potential in the US market where willingness to pay is structurally higher. For instance, Doosan Robotics trades at ~23x 2028E P/S despite being focused on collaborative robots rather than humanoids, vs. UBTECH and Dobot Robotics at ~4-6x P/S. Unitree, IPOed on August 19, is currently trading at ~30x P/S, likely reflecting a post-IPO honeymoon effect and A-share scarcity issues. That said, investors also appear to be underwriting Unitree's broader application runway more than peers, given the company's strategic push across multiple robot form factors and use cases.
- Valuation dispersion across the broader robotics supply chain is wide. Market leaders trade at ~5-31x 2028E P/S, driven by differences in margin structure and where investors believe the key bottlenecks sit (notably reducers and actuators).
- Conventional industrial robot OEMs typically trade at ~2-6x 2028E P/S as the sector is more mature and growth is closer to saturation. We include these as reference points, but view the comparison as less relevant given the structurally higher growth runway we expect for humanoids.

Table 6: Peer valuation comparison table

Company	Ticker	Mkt. cap (US\$bn)	P/E (x)		P/B (x)		P/S (x)		OPM		Rev growth	
			27E	28E	27E	28E	27E	28E	27E	28E	27E	28E
Hyundai Motor	005380 KS	61.5	9	8	0.8	0.7	0.4	0.4	7%	7%	3%	5%
Kia	000270 KS	37.0	5	5	0.7	0.6	0.4	0.4	9%	9%	9%	5%
Humanod/ Collaborative robot OEMs												
Doosan Robotics	454910 KS	3.3	n/a	n/a	15.4	15.1	47.3	23.2	-21%	-2%	46%	56%
Unitree	688836 CH	40.5	221	105	59.3	37.9	54.8	30.1	27%	28%	n/a	70%
UBTech	9880 HK	5.4	377	51	4.8	4.2	6.3	4.0	2%	10%	98%	51%
Dobot Robotics	2432 HK	1.3	n/a	255	3.9	3.8	8.1	5.6	-5%	14%	49%	42%
Robotics supply chain market leaders												
Harmonic Drive Systems	6324 JP	3.5	82	56	6.4	5.9	7.4	6.4	12%	16%	6%	28%
Leader Harmonic Drive	688017 CH	8.6	198	136	14.7	13.6	43.1	30.6	22%	24%	58%	50%
Orbbec	688322 CH	6.2	68	43	9.7	8.1	15.7	10.9	23%	25%	64%	68%
Haozhi Industrial	300503 CH	3.2	34	27	9.9	7.8	6.3	5.1	21%	22%	74%	32%
Zhaowei Machinery & Electronic	003021 CH	2.9	50	42	3.2	3.1	8.3	6.8	15%	14%	5%	17%
Fortior Technology	688279 CH	2.8	43	32	3.5	3.1	13.3	10.2	37%	38%	33%	31%
Traditional industrial robot OEMs												
ABB	ABBN SW	182.5	28	24	8.1	6.8	4.3	3.9	20%	21%	12%	12%
Fanuc	6954 JT	37.3	27	26	2.9	2.8	6.1	5.8	24%	24%	7%	14%
Yaskawa	6506 JT	8.4	27	22	2.5	2.4	2.2	2.1	10%	12%	-1%	12%
Innovance	300124 CH	23.8	22	18	3.5	3.0	2.5	2.2	12%	12%	18%	17%
Traditional auto parts suppliers expanding into actuators												
Schaeffler	SHA0 GR	8.4	14	8	2.5	2.1	0.3	0.3	5%	7%	-1%	4%
Hyundai Mobis	012330 KS	32.8	9	9	0.8	0.7	0.7	0.6	6%	6%	6%	6%
Mando	204320 KS	1.7	10	9	0.8	0.7	0.2	0.2	4%	5%	4%	6%
Sanhua Intelligent Controls	002050 CH	22.0	29	25	3.9	3.5	3.7	3.2	16%	16%	6%	16%
Hengli Hydraulic	601100 CH	21.1	33	28	6.2	5.4	8.9	7.6	29%	29%	26%	18%
Shuanghuan Driveline	002472 CH	4.6	18	15	2.4	2.1	2.6	2.3	17%	17%	9%	15%

Based on Bloomberg consensus estimates. Based on 21 Aug 2026 prices
Source: Bloomberg Finance L.P.

Table 7: HMG robotics business: estimated value

Wbn, units

	Value	Note
2030E Sales revenue	5,220	
ASP (\$/unit)	120,000	ASP by 2030E when Atlas enters initial mass-production
2030E shipment volume	30,000	30,000 units of Atlas humanoid
Target P/S multiple	20x	
Est. value (2030E)	104,400	
Est. value (2028E)	83,227	Using 12% discount rate to bring to its 2028 value; equivalent to \$57bn

Source: Company data, J.P. Morgan estimates

Table 8: HMC: SOTP valuation

Wbn, x, %

		Value	Note
Auto business	(a)	106,377	
2028E Net profit		11,820	
Target P/E		9x	Referenced by non-traditional OEMs with ADAS capabilities
Robotics business	(b) = 28%x(c)x(1-d)	15,750	28% stake in Hyundai Motor Group's robotics business
Estimated value (2028E)	(c)	80,357	Using 12% discount rate to 2030E est. value of W104tn
Holding company discount	(d)	30%	
Preferred shares	(e)	11,901	
Target common share market cap	(a)+(b)-(e)	110,226	
Price target		540,000	

Source: Bloomberg Finance L.P., J.P. Morgan estimates

Table 9: Kia: SOTP valuation

Wbn, x, %

		Value	Note
Auto business	(a)	70,614	
2028E Net profit		10,088	
Target P/E		7x	Referenced by non-traditional OEMs with ADAS capabilities; ~15% discount to HMC
Robotics business	(b) = 17%x(c)x(1-d)	9,562	17% stake in Hyundai Motor Group's robotics business
Estimated value (2028E)	(c)	80,357	Using 12% discount rate to 2030E est. value of W104tn
Holding company discount	(d)	30%	
Target market cap	(a)+(b)	80,176	
Price target		210,000	

Source: Bloomberg Finance L.P., J.P. Morgan estimates

Additional value justification (revenue and valuation)

Our current framework largely reflects the hardware monetization of the business. We have not incorporated several potential revenue streams, including: 1) non-humanoid robots (more task-specific); 2) a “data factory” business; and 3) after-sales services, including repairs and incremental monetization of OTA updates. From a valuation perspective, there is also upside optionality from: 1) a more restrictive geopolitical environment that limits low-cost Chinese competition; and 2) a shift from one-time hardware sales toward RaaS (subscription-based) pricing, which can improve cash flow visibility and support higher multiples.

- **Task-specific robots**, as discussed, may inflect earlier than humanoids given clearer ROI and lower cost for narrowly defined tasks with limited edge cases. We have not included Spot (quadruped) and Stretch (one-armed logistics robot) in our valuation due to limited disclosure, but adoption appears to be progressing, with ~1,500 Spot units already deployed and Stretch supported by a >1,000-unit global deployment MOU with logistics customers such as DHL.
- **The data-factory concept** is less familiar in traditional manufacturing. It refers to producing high-quality, real-world interaction data at scale to train and improve robotic AI models. Potential customers include robot OEMs, AI model developers, and industrial end users seeking to train robots for specific tasks.
- **Repair services and OTA updates** may be bundled at initial sale, but can become a recurring revenue stream post-warranty.
- **Geopolitics could also be a net tailwind.** Chinese robot OEMs remain the most credible competitive threat given their cost leadership and ambition, and the US is a key market with higher willingness to pay. That said, US regulatory frameworks such as the America Robotics Act appear to restrict Chinese robots, which we discussed in our note [here](#).
- **Shifting from one-time sales to RaaS (subscription-based revenue)** should improve cash flow stability and earnings visibility.

Overweight

005380.KS, 005380 KS

Price (24 Aug 26):W413,000

▲ Price Target (Dec-27):W540,000

Prior (Dec-27):W530,000

Korea Auto, Robotics, Battery and Utility

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Quarterly Forecasts (FYE Dec)

Adj. EPS (W)

	2025A	2026E	2027E
Q1	12,130	8,908A	7,746
Q2	11,365	9,615	11,282
Q3	8,572	10,400	12,646
Q4	3,918	6,919	9,124
FY	35,964	36,180	41,235

Style Exposure

Quant Factors	Current	Hist %Rank (1=Top)			
	%Rank	6M	1Y	3Y	5Y
Value	9	11	3	3	28
Growth	83	80	78	80	47
Momentum	54	13	78	24	69
Quality	55	49	50	61	65
Low Vol	54	43	26	12	52
ESGQ	93	100	100	98	99

Hyundai Motor Company

Direct beneficiary of HMG's robotics business; raise PT to W540,000

We maintain Overweight and raise our Dec-27 PT on HMC to W540,000 (from W530,000), while keeping our core estimates largely unchanged. The increase reflects a higher robotics business value in our SOTP (\$57bn from \$49bn). We value the core auto business at W106tn, applying a 9x target P/E to 2028E net profit of W11.8tn, benchmarked to non-traditional OEMs with ADAS capabilities. We also ascribe W16tn to HMC's 28% stake in Hyundai Motor Group's robotics business, based on a 2030E value of W104tn discounted at 12% and applying a 30% holding company discount. We believe auto earnings are largely de-risked and investor focus should increasingly shift to HMG's robotics optionality, where HMC remains the most direct beneficiary.

Price Performance



	YTD	1m	3m	12m
Abs	39.1%	3.1%	-37.0%	87.7%
Rel	-19.8%	3.0%	-22.3%	-20.9%

Company Data

Shares O/S (mn)	260
52-week range (W)	787,000-211,000
Market cap (\$ mn)	77,572
Exchange rate	1,385.83
Free float (%)	57.8%
3M ADV (mn)	2.22
3M ADV (\$ mn)	903.0
Volatility (90 Day)	77
Index	KOSPI
BBG ANR (Buy Hold Sell)	31 5 0

Key Metrics (FYE Dec)

W in billions	FY25A	FY26E	FY27E	FY28E
Financial Estimates				
Revenue	186,254	196,121	207,321	213,290
Adj. EBITDA	16,484	17,661	22,239	23,924
Adj. EBIT	11,468	11,211	13,065	14,165
Adj. net income	9,446	9,376	10,540	11,820
Net margin	5.1%	4.8%	5.1%	5.5%
Adj. EPS	35,964	36,180	41,235	46,239
BBG EPS	44,979	42,131	47,718	52,465
Cashflow from operations	(5,991)	21,459	17,484	20,822
FCFF	(14,358)	12,459	7,984	10,822
Margins and Growth				
Revenue Growth Y/Y (%)	6.3%	5.3%	5.7%	2.9%
EBITDA margin	8.9%	9.0%	10.7%	11.2%
EBITDA Growth Y/Y (%)	(11.0%)	7.1%	25.9%	7.6%
EBIT margin	6.2%	5.7%	6.3%	6.6%
Adj. EPS growth	(24.3%)	0.6%	14.0%	12.1%
Ratios				
Adj. tax rate	25.1%	24.8%	27.4%	26.4%
Interest cover	NM	NM	NM	NM
Net debt/Equity	1.2	1.1	1.0	0.9
Net debt/EBITDA	9.0	8.1	6.4	5.7
ROCE	3.2%	3.0%	3.2%	3.5%
ROE	8.4%	7.9%	8.5%	8.9%
Valuation				
FCFF yield	(13.2%)	11.6%	7.6%	10.3%
Dividend yield	2.4%	2.4%	2.8%	3.1%
EV/EBITDA	16.3	15.0	11.9	10.9
EV/Revenue	1.4	1.3	1.3	1.2
Adj. P/E	11.5	11.4	10.0	8.9

Summary Investment Thesis and Valuation

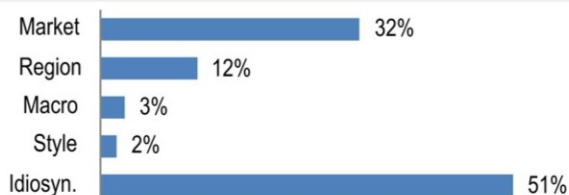
Investment Thesis

We remain Overweight on HMC. Near-term, US HEV-led market share gains continue to support auto operations. Longer-term, we see robotics as the key narrative driver as investor focus shifts toward tangible progress in humanoids. We refresh our humanoid ASP framework and continue to view P/S as the more relevant valuation method at this stage, as early deployments may price close to COGS/BOM and remain loss-making during the ramp.

Valuation

Our Dec-27 PT of W540,000 is based on SOTP: 1) valuing the auto business at 9x 2028E P/E (referenced by non-traditional OEMs with ADAS capabilities); and 2) including the value of a 28% stake in Hyundai Motor Group's robotics business at ~W16tn, which reflects the mid-point of our bull/bear analysis after applying a 30% holding company discount.

Performance Drivers



Factors	6M Corr	1Y Corr
Market: MSCI Asia Pac ex JP	0.68	0.56
Region: Korea	0.22	0.42
Macro:		
JPM China A-shares Sentiment	-0.35	-0.24
Generic 1st 'CO' Future	0.42	0.19
JPM Forecast Revision EM	-0.23	0.09
Quant Styles:		
Growth	-0.10	-0.12
LowVol	0.05	0.08
Value	0.14	0.07

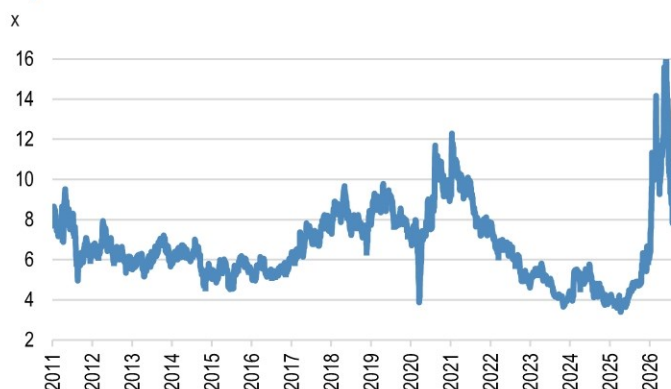
Table 10: HMC: Summary of key assumptions and financials

'000 units (retail sales), Won (FX), US\$ (Incentives), Wbn (Financials)

	2023	2024	2025	2026E	2027E	2028E	2029E	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E
Retail sales	4,157	4,047	4,094	4,092	4,293	4,412	4,493	956	1,043	1,036	1,059	949	999	1,034	1,110
Korea	762	708	713	642	706	727	742	166	189	181	178	159	158	152	173
US	870	912	982	1,012	1,042	1,073	1,095	221	256	261	245	224	266	260	262
Europe	559	542	533	507	532	532	532	133	142	133	125	130	130	124	123
India	595	595	568	625	675	708	723	137	131	137	162	153	141	152	178
FX															
USDKRW	1,306	1,365	1,423	1,492	1,500	1,500	1,500	1,453	1,401	1,387	1,452	1,467	1,502	1,500	1,500
Incentives															
US incentives	1,840	2,913	3,130	2,883	3,200	3,200	3,200	3,198	2,959	3,634	2,729	2,508	3,025	3,000	3,000
Financials															
Revenue	162,701	175,232	186,255	196,121	207,321	213,290	217,304	44,408	48,287	46,721	46,839	45,939	49,215	49,530	51,436
Auto	130,186	136,725	145,053	151,996	162,182	166,956	170,297	34,718	37,030	36,715	36,590	34,539	36,832	39,558	41,067
Finance	22,401	28,446	30,288	33,148	33,383	34,043	34,715	7,398	8,269	7,189	7,432	8,991	9,504	7,206	7,446
Others	10,114	10,061	10,914	10,977	11,756	12,292	12,292	2,292	2,988	2,817	2,817	2,409	2,879	2,766	2,923
% y/y	14%	8%	6%	5%	6%	3%	2%	9%	7%	9%	0%	3%	2%	6%	10%
OP	15,127	14,240	11,468	11,211	13,065	14,165	15,030	3,634	3,602	2,537	1,695	2,515	2,851	3,216	2,629
% y/y	54%	-6%	-19%	-2%	17%	8%	6%	2%	-16%	-29%	-40%	-31%	-21%	27%	55%
OP margin	9.3%	8.1%	6.2%	5.7%	6.3%	6.6%	6.9%	8.2%	7.5%	5.4%	3.6%	5.5%	5.8%	6.5%	5.1%
NP	12,272	13,230	10,365	10,458	11,553	12,749	13,656	3,382	3,250	2,548	1,184	2,585	2,888	2,945	2,040
% y/y	54%	8%	-22%	1%	10%	10%	7%	0%	-22%	-21%	-52%	-24%	-11%	16%	72%
NP margin	7.5%	7.5%	5.6%	5.3%	5.6%	6.0%	6.3%	7.6%	6.7%	5.5%	2.5%	5.6%	5.9%	5.9%	4.0%
NP to parent	11,962	12,527	9,446	9,376	10,540	11,820	12,720	3,157	2,998	2,261	1,029	2,335	2,521	2,726	1,793

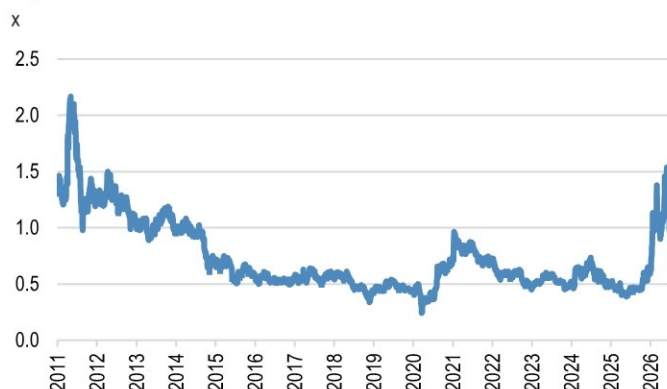
Source: Company data, Bloomberg Finance L.P., J.P. Morgan estimates

Figure 13: HMC: 12m forward P/E bands



Source: Bloomberg Finance L.P.

Figure 14: HMC: 12m forward P/B bands



Source: Bloomberg Finance L.P.

Investment Thesis, Valuation and Risks

Hyundai Motor Company (Overweight; Price Target: W540,000)

Investment Thesis

We remain Overweight on HMC. Near-term, US HEV-led market share gains continue to support auto operations. Longer-term, we see robotics as the key narrative driver as investor focus shifts toward tangible progress in humanoids. We refresh our humanoid ASP framework and continue to view P/S as the more relevant valuation method at this stage, as early deployments may price close to COGS/BOM and remain loss-making during the ramp.

Valuation

Our Dec-27 PT of W540,000 is based on SOTP: 1) valuing the auto business at 9x 2028E P/E (referenced by non-traditional OEMs with ADAS capabilities); and 2) including the value of a 28% stake in Hyundai Motor Group's robotics business at ~W16tn, which reflects the mid-point of our bull/bear analysis after applying a 30% holding company discount.

HMC: SOTP valuation

Wbn, x, %

		Value	Note
Auto business	(a)	106,377	
2028E Net profit		11,820	
Target P/E		9x	Referenced by non-traditional OEMs with ADAS capabilities
Robotics business	(b) = 28%x(c)x(1-d)	15,750	28% stake in Hyundai Motor Group's robotics business
Estimated value (2028E)	(c)	80,357	Using 12% discount rate to 2030E est. value of W104tn
Holding company discount	(d)	30%	
Preferred shares	(e)	11,901	
Target common share market cap	(a)+(b)-(e)	110,226	
Price target		540,000	

Source: Bloomberg Finance L.P., J.P. Morgan estimates.

Risks to Rating and Price Target

Risks to our rating and price target include: 1) heightened macro uncertainties exerting downward pressure on auto sales volumes; 2) delays in achieving key development milestones for the ADAS system; and 3) possible setbacks in HMG's robotics development and manufacturing plans.

Overweight

000270.KS, 000270 KS

Price (24 Aug 26): W131,200

▲ **Price Target (Dec-27): W210,000**

Prior (Dec-27): W200,000

Korea Auto, Robotics, Battery and Utility

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J.P. Morgan Securities (Far East) Limited, Seoul Branch

Quarterly Forecasts (FYE Dec)

Adj. EPS (W)

	2025A	2026E	2027E
Q1	6,087	4,713A	4,704
Q2	5,823	5,990	5,926
Q3	3,685	6,310	7,431
Q4	3,794	5,245	5,565
FY	19,457	22,258	23,625

Style Exposure

Quant Factors	Current	Hist %Rank (1=Top)			
	%Rank	6M	1Y	3Y	5Y
Value	11	14	11	10	17
Growth	13	40	63	45	38
Momentum	61	48	81	25	31
Quality	4	8	18	22	27
Low Vol	35	21	21	24	66
ESGQ	99	98	94	94	97

Kia Corp

Underappreciated way to gain exposure to the HMG robotics story; raise PT to W210,000

We maintain Overweight and raise our Dec-27 PT on Kia to W210,000 (from W200,000), while keeping our core estimates unchanged. The increase reflects a modest uplift in the Group's robotics business value (\$57bn from \$49bn) in our SOTP. We value the core auto business at W71tn, applying a 7x target P/E to 2028E net profit of W10.1tn, benchmarked to non-traditional OEMs with ADAS capabilities and applying a ~15% discount vs. HMC. We also ascribe W10tn to Kia's 17% stake in Hyundai Motor Group's robotics business, based on a 2030E value of W104tn discounted at 12% and applying a 30% holding company discount.

Price Performance



	YTD	1m	3m	12m
Abs	7.9%	-0.8%	-20.3%	25.2%
Rel	-51.0%	-0.9%	-5.6%	-83.4%

Company Data

Shares O/S (mn)	394
52-week range (W)	212,500-98,400
Market cap (\$ mn)	37,273
Exchange rate	1,385.83
Free float (%)	52.9%
3M ADV (mn)	1.49
3M ADV (\$ mn)	161.8
Volatility (90 Day)	59
Index	KOSPI
BBG ANR (Buy Hold Sell)	34 0 0

Key Metrics (FYE Dec)

W in billions	FY25A	FY26E	FY27E	FY28E
Financial Estimates				
Revenue	114,141	128,144	133,241	135,609
Adj. EBITDA	11,792	13,706	15,367	16,680
Adj. EBIT	9,078	10,232	10,669	11,608
Adj. net income	7,561	8,650	9,181	10,088
Net margin	6.6%	6.7%	6.9%	7.4%
Adj. EPS	19,457	22,258	23,625	25,959
BBG EPS	19,473	21,918	24,040	25,951
Cashflow from operations	9,054	13,539	13,795	15,093
FCFF	4,061	7,339	7,295	7,793
Margins and Growth				
Revenue Growth Y/Y (%)	6.2%	12.3%	4.0%	1.8%
EBITDA margin	10.3%	10.7%	11.5%	12.3%
EBITDA Growth Y/Y (%)	(22.5%)	16.2%	12.1%	8.5%
EBIT margin	8.0%	8.0%	8.0%	8.6%
Adj. EPS growth	(21.6%)	14.4%	6.1%	9.9%
Ratios				
Adj. tax rate	26.2%	26.5%	26.0%	26.0%
Interest cover	NM	NM	NM	NM
Net debt/Equity	NM	NM	NM	NM
Net debt/EBITDA	NM	NM	NM	NM
ROCE	10.9%	11.4%	11.0%	11.0%
ROE	12.9%	13.5%	13.2%	13.3%
Valuation				
FCFF yield	8.0%	14.4%	14.3%	15.3%
Dividend yield	5.2%	5.2%	5.3%	5.9%
EV/EBITDA	2.3	1.7	1.3	0.9
EV/Revenue	0.2	0.2	0.1	0.1
Adj. P/E	6.7	5.9	5.6	5.1

Summary Investment Thesis and Valuation

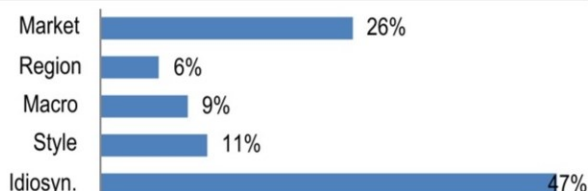
Investment Thesis

We maintain our Overweight rating on Kia as a beneficiary of both a stabilizing auto earnings base and HMG's robotics optionality. US HEV-led market share gains provide a near-term earnings floor, while Kia's 17% stake in HMG's robotics business offers meaningful exposure. We view Kia as an attractive, relatively underappreciated way to gain exposure to the HMG robotics story at a valuation discount to HMC.

Valuation

Our Dec-27 PT of W210,000 is based on SOTP: 1) valuing the auto business at 7x 2028E P/E (referenced by non-traditional OEMs with ADAS capabilities); and 2) including the value of a 17% stake in Hyundai Motor Group's robotics business at ~W8tn, which reflects the mid-point of our bull/bear analysis after applying a 30% holding company discount.

Performance Drivers



Factors	6M Corr	1Y Corr
Market: MSCI Asia Pac ex JP	0.56	0.51
Region: Korea	0.11	0.28
Macro:		
JPM EM Currency(EMCI) Fixing	0.27	0.23
Emerging Economies CPI(YoY)	0.29	0.19
JP Morgan GBI-EM Global Div	0.22	0.19
Quant Styles:		
Size	-0.29	-0.25
Growth	-0.32	-0.25
LowVol	0.31	0.21

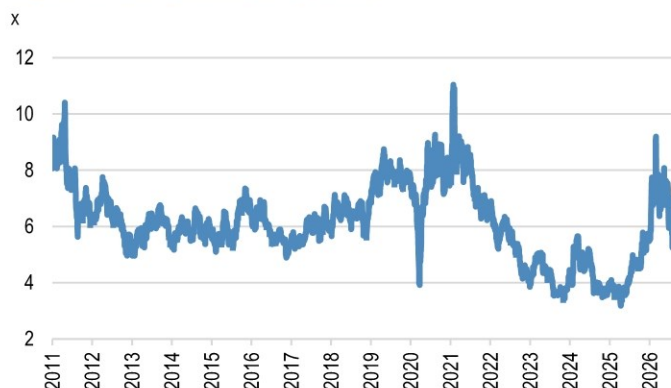
Table 11: Kia: Summary of key assumptions and financials

'000 units (retail sales), Won (FX), US\$ (incentives), Wbn (financials)

	2023	2024	2025	2026E	2027E	2028E	2029E	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E
Retail sales	3,012	2,982	3,091	3,286	3,361	3,416	3,461	752	791	774	774	779	837	818	853
Korea	564	540	546	589	572	572	577	134	142	138	132	142	154	141	152
US	782	796	852	886	913	940	959	199	218	220	216	207	224	229	227
Europe	602	559	536	585	596	584	584	145	142	131	119	146	159	145	136
India	245	243	280	328	361	379	379	69	63	62	86	77	79	77	94
FX															
USDKRW	1,306	1,365	1,423	1,492	1,500	1,500	1,500	1,453	1,401	1,387	1,452	1,467	1,502	1,500	1,500
Incentives															
US incentives	1,168	3,104	3,264	3,285	3,200	3,200	3,200	3,243	2,874	3,450	3,489	3,382	3,357	3,200	3,200
Financials															
Revenue	99,808	107,449	114,141	128,144	133,241	135,609	137,575	28,018	29,350	28,686	28,088	29,502	33,037	33,258	32,347
% y/y	15%	8%	6%	12%	4%	2%	1%	7%	6%	8%	3%	5%	13%	16%	15%
OP	11,608	12,667	9,078	10,232	10,669	11,608	12,042	3,009	2,765	1,462	1,843	2,205	2,629	2,917	2,481
% y/y	60%	9%	-28%	13%	4%	9%	4%	-12%	-24%	-49%	-32%	-27%	-5%	99%	35%
OP margin	11.6%	11.8%	8.0%	8.0%	8.0%	8.6%	8.8%	10.7%	9.4%	5.1%	6.6%	7.5%	8.0%	8.8%	7.7%
NP	8,778	9,775	7,554	8,649	9,181	10,088	10,621	2,393	2,268	1,422	1,471	1,830	2,328	2,452	2,038
% y/y	62%	11%	-23%	14%	6%	10%	5%	-15%	-23%	-37%	-16%	-24%	3%	72%	39%
NP margin	8.8%	9.1%	6.6%	6.7%	6.9%	7.4%	7.7%	8.5%	7.7%	5.0%	5.2%	6.2%	7.0%	7.4%	6.3%
NP to parent	8,777	9,773	7,561	8,650	9,181	10,088	10,621	2,393	2,269	1,425	1,474	1,831	2,328	2,452	2,038

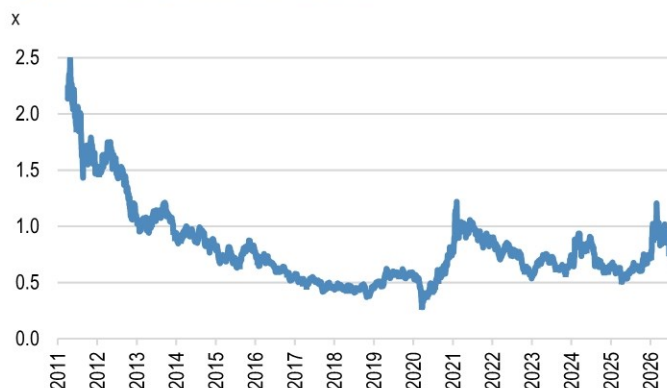
Source: Company data, Bloomberg Finance L.P. and J.P. Morgan estimates.

Figure 15: Kia: 12m forward P/E bands



Source: Bloomberg Finance L.P.

Figure 16: Kia: 12m forward P/B bands



Source: Bloomberg Finance L.P.

Investment Thesis, Valuation and Risks

Kia Corp (Overweight; Price Target: W210,000)

Investment Thesis

We maintain our Overweight rating on Kia as a beneficiary of both a stabilizing auto earnings base and HMG's robotics optionality. US HEV-led market share gains provide a near-term earnings floor, while Kia's 17% stake in HMG's robotics business offers meaningful exposure. We view Kia as an attractive, relatively underappreciated way to gain exposure to the HMG robotics story at a valuation discount to HMC.

Valuation

Our Dec-27 PT of W210,000 is based on SOTP: 1) valuing the auto business at 7x 2028E P/E (referenced by non-traditional OEMs with ADAS capabilities); and 2) including the value of a 17% stake in Hyundai Motor Group's robotics business at ~W8tn, which reflects the mid-point of our bull/bear analysis after applying a 30% holding company discount.

Kia: SOTP valuation

Wbn, x, %

		Value	Note
Auto business	(a)	70,614	
2028E Net profit		10,088	
Target P/E		7x	Referenced by non-traditional OEMs with ADAS capabilities; ~15% discount to HMC
Robotics business	(b) = 17%x(c)x(1-d)	9,562	17% stake in Hyundai Motor Group's robotics business
Estimated value (2028E)	(c)	80,357	Using 12% discount rate to 2030E est. value of W104tn
Holding company discount	(d)	30%	
Target market cap	(a)+(b)	80,176	
Price target		210,000	

Source: Bloomberg Finance L.P., J.P. Morgan estimates

Risks to Rating and Price Target

Risks to our rating and price target include: 1) heightened macro uncertainties exerting downward pressure on auto sales volumes; 2) delays in achieving key development milestones for the ADAS system; and 3) possible setbacks in HMG's robotics development and manufacturing plans.

Kia Corp: Summary of Financials

Income Statement	FY25A	FY26E	FY27E	FY28E	Cash Flow Statement	FY25A	FY26E	FY27E	FY28E
Revenue	114,141	128,144	133,241	135,609	Cash flow from operating activities	9,054	13,539	13,795	15,093
COGS	(91,633)	(103,057)	(106,548)	(107,692)	o/w Depreciation & amortization	2,714	3,473	4,698	5,072
Gross profit	22,508	25,086	26,693	27,917	o/w Changes in working capital	(615)	1,417	(84)	(66)
SG&A	(13,430)	(14,854)	(16,025)	(16,308)	Cash flow from investing activities	(4,960)	(6,462)	(6,500)	(7,300)
Adj. EBITDA	11,792	13,706	15,367	16,680	o/w Capital expenditure	(3,765)	(4,767)	(5,033)	(5,800)
D&A	(2,714)	(3,473)	(4,698)	(5,072)	as % of sales	3.3%	3.7%	3.8%	4.3%
Adj. EBIT	9,078	10,232	10,669	11,608	Cash flow from financing activities	(3,663)	(3,541)	(3,329)	(2,720)
Net Interest	874	807	899	1,055	o/w Dividends paid	(2,559)	(2,642)	(2,642)	(2,720)
Adj. PBT	10,241	11,771	12,407	13,632	o/w Shares issued/(repurchased)	(689)	(687)	(687)	0
Tax	(2,687)	(3,123)	(3,226)	(3,544)	o/w Net debt issued/(repaid)	(890)	(211)	0	0
Minority Interest	7	1	0	0	Net change in cash	432	3,536	3,966	5,073
Adj. Net Income	7,561	8,650	9,181	10,088	Adj. Free cash flow to firm	4,061	7,339	7,295	7,793
Reported EPS	19,457	22,258	23,625	25,959	y/y Growth	(48.5%)	80.7%	(0.6%)	6.8%
Adj. EPS	19,457	22,258	23,625	25,959					
DPS	6,800	6,800	7,000	7,800					
Payout ratio	34.9%	30.6%	29.6%	30.0%					
Shares outstanding	389	389	389	389					
Balance Sheet	FY25A	FY26E	FY27E	FY28E	Ratio Analysis	FY25A	FY26E	FY27E	FY28E
Cash and cash equivalents	13,998	17,535	21,501	26,574	Gross margin	19.7%	19.6%	20.0%	20.6%
Accounts receivable	6,963	7,119	7,402	7,534	EBITDA margin	10.3%	10.7%	11.5%	12.3%
Inventories	14,666	14,722	15,221	15,385	EBIT margin	8.0%	8.0%	8.0%	8.6%
Other current assets	22,190	22,402	23,184	23,479	Net profit margin	6.6%	6.7%	6.9%	7.4%
Current assets	44,426	48,437	53,185	58,553	ROE	12.9%	13.5%	13.2%	13.3%
PP&E	19,934	22,071	23,336	25,087	ROA	7.9%	8.5%	8.4%	8.7%
LT investments	28,149	28,149	28,149	28,149	ROCE	10.9%	11.4%	11.0%	11.0%
Other non current assets	29,813	29,813	29,813	29,813	SG&A/Sales	11.8%	11.6%	12.0%	12.0%
Total assets	98,979	105,717	112,267	119,863	Net debt/Equity	NM	NM	NM	NM
Short term borrowings	1,365	1,154	1,154	1,154	Net debt/EBITDA	NM	NM	NM	NM
Payables	18,982	20,611	21,310	21,538	Sales/Assets (x)	1.2	1.3	1.2	1.2
Other short term liabilities	8,032	8,032	8,032	8,032	Assets/Equity (x)	1.6	1.6	1.6	1.5
Current liabilities	28,378	29,797	30,495	30,724	Interest cover (x)	NM	NM	NM	NM
Long-term debt	979	979	979	979	Operating leverage	(454.9%)	103.6%	107.2%	495.8%
Other long term liabilities	9,410	9,410	9,410	9,410	Tax rate	26.2%	26.5%	26.0%	26.0%
Total liabilities	37,789	39,207	39,905	40,134	Revenue y/y Growth	6.2%	12.3%	4.0%	1.8%
Shareholders' equity	61,188	66,508	72,360	79,727	EBITDA y/y Growth	(22.5%)	16.2%	12.1%	8.5%
Minority interests	3	2	2	2	EPS y/y Growth	(21.6%)	14.4%	6.1%	9.9%
Total liabilities & equity	98,979	105,717	112,267	119,863	Valuation	FY25A	FY26E	FY27E	FY28E
BVPS	157,456	171,147	186,205	205,164	P/E (x)	6.7	5.9	5.6	5.1
y/y Growth	11.0%	8.7%	8.8%	10.2%	P/BV (x)	0.8	0.8	0.7	0.6
Net debt/(cash)	(19,892)	(23,902)	(27,868)	(32,941)	EV/EBITDA (x)	2.3	1.7	1.3	0.9
					Dividend Yield	5.2%	5.2%	5.3%	5.9%

Source: Company reports and J.P. Morgan estimates.

Note: W in billions (except per-share data). Fiscal year ends Dec. o/w - out of which

Companies Discussed in This Report (all prices in this report as of market close on 24 August 2026, unless otherwise indicated)
HL Mando(204320.KS/W51,100/N), Hyundai Autoever(307950.KS/W457,000/N), Hyundai Mobis(012330.KS/W498,000/OW)

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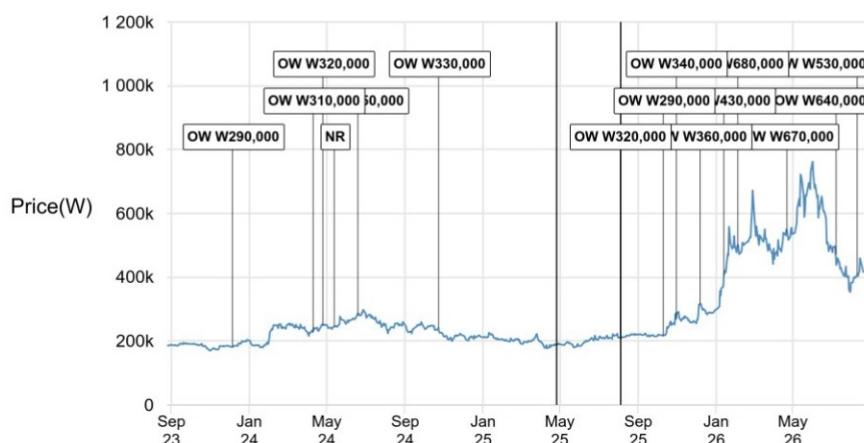
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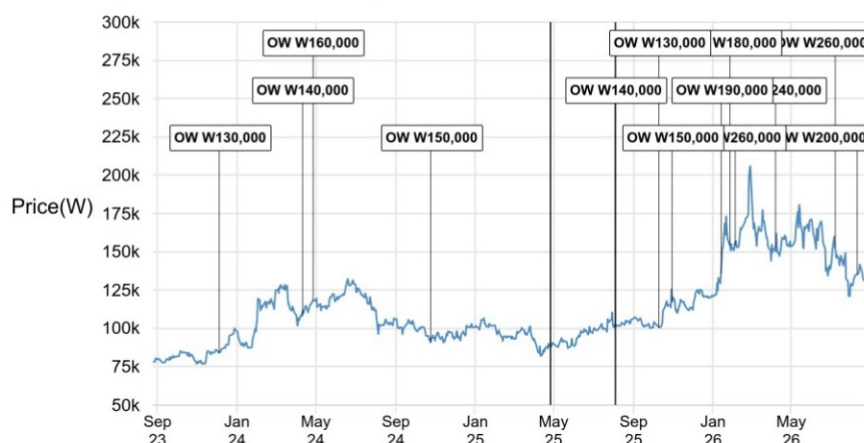
Hyundai Motor Company (005380.KS, 005380 KS) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Jun 14, 2000. All share prices are as of market close on the previous business day.
Break in coverage Apr 25, 2025 - Aug 04, 2025.

Date	Rating	Price (W)	Price Target (W)
06-Dec-23	OW	181400	290,000
11-Apr-24	OW	228000	310,000
26-Apr-24	OW	250000	320,000
14-May-24	NR	245000	--
19-Jun-24	OW	283000	350,000
24-Oct-24	OW	241000	330,000
04-Aug-25	OW	210000	320,000
11-Oct-25	OW	217000	290,000
31-Oct-25	OW	269000	340,000
08-Dec-25	OW	314500	360,000
14-Jan-26	OW	406500	430,000
04-Feb-26	OW	494500	680,000
23-Apr-26	OW	540000	670,000
08-Jul-26	OW	471000	640,000
11-Aug-26	OW	409000	530,000

Kia Corp (000270.KS, 000270 KS) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Jun 14, 2000. All share prices are as of market close on the previous business day.
Break in coverage Apr 25, 2025 - Aug 04, 2025.

Date	Rating	Price (W)	Price Target (W)
06-Dec-23	OW	84100	130,000
11-Apr-24	OW	108000	140,000
27-Apr-24	OW	118200	160,000
25-Oct-24	OW	90900	150,000
04-Aug-25	OW	100800	140,000
11-Oct-25	OW	100600	130,000
31-Oct-25	OW	117300	150,000
14-Jan-26	OW	135900	190,000
28-Jan-26	OW	154200	180,000
04-Feb-26	OW	154300	260,000
07-Apr-26	OW	153000	240,000
08-Jul-26	OW	152700	260,000
11-Aug-26	OW	135200	200,000

HL Mando (204320.KS, 204320 KS) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Oct 07, 2014. All share prices are as of market close on the previous business day.
Break in coverage Apr 25, 2025 - Jun 19, 2026.

Date	Rating	Price (W)	Price Target (W)
27-Oct-23	OW	35700	52,000
06-Dec-23	OW	34900	50,000
06-Feb-24	OW	36850	45,000
11-Jun-24	N	46750	50,000
26-Jul-24	OW	39000	50,000
19-Jun-26	UW	69200	50,000
07-Jul-26	N	49700	50,000

Hyundai Autoever (307950.KS, 307950 KS) Price Chart



Date	Rating	Price (W)	Price Target (W)
11-Sep-23	N	217500	210,000
27-Oct-23	N	158400	180,000
27-Jan-24	UW	178200	140,000
30-Apr-24	UW	161400	130,000
26-Jul-24	UW	154300	140,000
30-Oct-24	UW	140100	120,000
03-Feb-25	UW	143000	110,000
28-Jul-25	UW	157100	130,000
31-Oct-25	UW	164700	150,000
30-Jan-26	UW	490000	160,000
19-Jun-26	N	667000	750,000
29-Jul-26	N	335000	390,000

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Apr 21, 2023. All share prices are as of market close on the previous business day.

Hyundai Mobis (012330.KS, 012330 KS) Price Chart



Date	Rating	Price (W)	Price Target (W)
28-Oct-23	OW	220000	300,000
06-Dec-23	OW	225000	310,000
26-Jan-24	OW	209500	280,000
26-Apr-24	OW	244000	300,000
25-Oct-24	OW	250000	330,000
19-Jun-26	N	606000	710,000
07-Jul-26	OW	512000	710,000

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage May 02, 2001. All share prices are as of market close on the previous business day.
Break in coverage Apr 25, 2025 - Jun 19, 2026.

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